

Sharp-constant Gaussian centroid dimension–accuracy tradeoffs

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Research note (7 October 2026, PDT). This is a strict continuation of the public all-integer Gaussian quadratic-dimension theorem and its spherical-cap converse. The analytic results below apply to arbitrary measurable, exactly equal-mass Gaussian partitions. We claim the proved statements, not historical priority over Gaussian vector quantization theory.

Abstract

Let

$$F_d(k) = \sup_{\substack{(A_i)_{i=1}^k \\ \gamma_d(A_i)=1/k}} \sum_{i=1}^k \left\| \int_{A_i} x d\gamma_d(x) \right\|^2, \quad F_\infty(k) = F_{k-1}(k).$$

Set $L = \log k$, $t_k = \Phi^{-1}(1 - 1/k)$, $h_k = k\varphi(t_k)$ and $U_k = h_k^2/k$.

We prove three improvements to the previously published fixed-additive-accuracy dimensional theorem:

1. **Sharp Gaussian extreme-value normalization and a universal asymptotic global gap.** Let γ denote the Euler–Mascheroni constant. Then

$$0 \leq \limsup_{k \rightarrow \infty} k(U_k - F_\infty(k)) \leq 2(1 - \gamma). \quad (1)$$

The rightmost constant comes from a dimension- $(k - 1)$ regular-simplex Gaussian partition; exact finite- k global simplex optimality is **not** assumed.

2. **Asymptotically sharp spherical-cap dimensional obstruction.** For every fixed $c > 0$, and any integer sequence d_k with $d_k/L^2 \rightarrow c$,

$$\liminf_{k \rightarrow \infty} k[U_k - F_{d_k}(k)] \geq \frac{1}{c}. \quad (2)$$

Consequently, for any fixed finite $C \geq 0$, if $D_C(k)$ denotes the least dimension achieving $F_d(k) \geq F_\infty(k) - C/k$, then

$$\liminf_{k \rightarrow \infty} \frac{D_C(k)}{(\log k)^2} \geq \frac{1}{C + 2(1 - \gamma)}. \quad (3)$$

This improves the previous quantitative coefficient $1/(C + 16)$ by a fully identified extreme-value constant.

3. **A dimension–accuracy continuum of exact-mass constructions.** For every fixed $A \geq 1$ and all sufficiently large *arbitrary* integers k , there is a partition in

$$d \leq \lceil A(\log k)^2 \rceil + 1$$

satisfying

$$U_k - F_d(k) \leq \frac{\mathcal{C}(A)}{k}, \quad \mathcal{C}(A) = 4 + 4 \log 2 + 2\sqrt{2} \left(5 + \frac{10}{\sqrt{A}} + \frac{8}{A} \right). \quad (4)$$

The proof refines the exponential tilting of the previous binary linear-code construction and uses the exact binary selector entropy bound $H(w) < 2 \log 2$. For example, the pairs

$$(A, C) = (1, 72), (4, 41), (16, 30), (64, 25), (256, 23), (1024, 22), (262144, 21) \quad (5)$$

are all rigorous integer-constant guarantees. In particular, $D_{21}(k) \leq \lceil 262144(\log k)^2 \rceil + 1$ for all sufficiently large k , whereas the older construction needed tolerance $92/k$ at coefficient $A = 1$.

The results improve concrete error guarantees and dimension coefficients; they do **not** identify the exact function $D_C(k)$, the best possible constant, or the fixed- k Standard Simplex optimizer.

1. Gaussian extremes through the constant term

Write $\bar{\Phi}(x) = \int_x^\infty \varphi(u) du$. Let $Z_1, Z_2, \dots$ be independent standard normals and $M_k = \max_{i \leq k} Z_i$. Write $m_k = \mathbb{E} M_k$.

Lemma 1 (two constant-term expansions). For $k \rightarrow \infty$, with $L = \log k$,

$$\begin{aligned} t_k^2 &= 2L - \log L - \log(4\pi) + o(1), \\ h_k^2 &= 2L - \log L - \log(4\pi) + 2 + o(1), \quad (6) \\ m_k^2 &= 2L - \log L - \log(4\pi) + 2\gamma + o(1). \end{aligned}$$

Here

$$\gamma = - \int_0^\infty e^{-u} \log u \, du$$

is Euler's constant. In particular,

$$h_k^2 - m_k^2 \rightarrow 2(1 - \gamma). \quad (7)$$

Proof. Twice integrating $\varphi'(x) = -x\varphi(x)$ by parts and using the elementary Mills upper bound give, for $t > 0$,

$$\bar{\Phi}(t) = \varphi(t) (t^{-1} - t^{-3} + O(t^{-5})). \quad (8)$$

The remainder is positive and bounded by $3\varphi(t)t^{-5}$; thus this is a rigorous asymptotic with an explicit tail estimate. Since $k\bar{\Phi}(t_k) = 1$, taking logarithms yields

$$L = t_k^2/2 + \log t_k + \frac{1}{2} \log(2\pi) + O(t_k^{-2}).$$

This first gives $t_k^2/(2L) \rightarrow 1$, and substitution back proves the first line of (6). Formula (8) gives $h_k = t_k + t_k^{-1} + O(t_k^{-3})$, so $h_k^2 = t_k^2 + 2 + o(1)$.

For the third line, define the standardized extreme

$$Y_k = t_k(M_k - t_k).$$

For each fixed real x ,

$$\Pr(Y_k \leq x) = [1 - \bar{\Phi}(t_k + x/t_k)]^k \rightarrow e^{-e^{-x}},$$

because the ratio of the two normal tails tends to e^{-x} by (8). We verify **uniform integrability**, rather than inferring expectation convergence from weak convergence.

For $x \geq 0$, the Gaussian upper-tail hazard $\varphi(t)/\bar{\Phi}(t) \geq t$ for $t > 0$ implies

$$\Pr(Y_k > x) \leq k\bar{\Phi}(t_k + x/t_k) \leq e^{-x}.$$

For $0 \leq x \leq t_k^2/2$, integrating the same hazard backwards gives

$$k\bar{\Phi}(t_k - x/t_k) \geq e^{x-x^2/(2t_k^2)} \geq e^{3x/4},$$

so

$$\Pr(Y_k < -x) \leq e^{-e^{3x/4}}.$$

For $t_k^2/2 \leq x \leq t_k^2$, monotonicity bounds the lower tail by $e^{-e^{3t_k^2/8}}$; its integral over this finite-length interval tends to zero. For $x \geq t_k^2$, substitute $x = t_k^2 + t_k u$, $u \geq 0$, and note

$$\int_{t_k^2}^{\infty} \Pr(Y_k < -x) dx = t_k \int_0^{\infty} \Phi(-u)^k du \leq \frac{t_k 2^{-(k-1)}}{\sqrt{2\pi}} \rightarrow 0.$$

The positive and negative tails are therefore uniformly integrable, and

$$\mathbb{E}Y_k \rightarrow - \int_0^{\infty} e^{-u} \log u du = \gamma.$$

Consequently $m_k = t_k + \gamma/t_k + o(t_k^{-1})$, and $m_k^2 = t_k^2 + 2\gamma + o(1)$, proving (6)–(7). $\square$

Theorem 2 (unrestricted global optimum: constant gap). For every $k \geq 3$, the regular-simplex Gaussian Voronoi partition in $\mathbb{R}^{k-1}$ has squared-centroid objective $m_k^2/(k-1)$. Consequently

$$0 \leq U_k - F_{\infty}(k) \leq \frac{h_k^2}{k} - \frac{m_k^2}{k-1} = \frac{2(1-\gamma) + o(1)}{k}. \quad (9)$$

Proof. Let $v_1, \dots, v_k \in \mathbb{R}^{k-1}$ be unit regular-simplex vertices, $\langle v_i, v_j \rangle = -1/(k-1)$ for $i \neq j$. Partition by the largest Gaussian score $\langle v_i, G \rangle$; the cells all have mass $1/k$. The Gaussian score vector has the same law as $\sqrt{k/(k-1)}(Z_i - \bar{Z})_{i=1}^k$, and the mean of its maximum is $\sqrt{k/(k-1)} m_k$. By simplex symmetry each cell centroid is collinear with its winning vertex. Writing $b_i = \int_{A_i} x d\gamma_{k-1}(x) = \beta v_i$, we get

$$k\beta = \mathbb{E} \max_i \langle v_i, G \rangle = \sqrt{k/(k-1)} m_k,$$

so the sum of squared centroid norms is $k\beta^2 = m_k^2/(k-1)$. The sharp one-cell Gaussian halfspace rearrangement gives $F_{\infty}(k) \leq U_k = h_k^2/k$. Finally $m_k^2 = O(L)$, so $km_k^2/(k-1) - m_k^2 = o(1)$, and (9) follows from Lemma 1. $\square$

Consequence (full constant-order global envelope). Without claiming convergence of the unknown global constant term, Theorem 2 establishes

$$2\gamma \leq \liminf_{k \rightarrow \infty} [kF_{\infty}(k) - 2\log k + \log \log k + \log(4\pi)] \leq \limsup_{k \rightarrow \infty} [kF_{\infty}(k) - 2\log k + \log \log k + \log(4\pi)] \leq 2. \quad (9a)$$

This replaces an unspecified $O(1)$ by the explicit classical constants 2γ and 2 .

2. A constant-sharp spherical-cap asymptotic

We recall the two analytical inequalities from the public spherical-cap converse, with short proofs.

Lemma 3 (centroid-to-spherical-maximum bridge). For every measurable equal-mass Gaussian k -partition in dimension $d \geq 3$, with objective P , there exist unit vectors $u_1, \dots, u_k \in S^{d-1}$ such that

$$\frac{kP}{h_k} \leq \mathbb{E} \max_i \langle u_i, G \rangle. \quad (10)$$

Proof. Let $b_i = \int_{A_i} x d\gamma_d(x)$. The halfspace rearrangement bounds $\|b_i\| \leq h_k/k$. Choose $u_i = b_i/\|b_i\|$, with any unit direction when $b_i = 0$. Pointwise, $\max_j \langle u_j, x \rangle \geq \sum_i 1_{A_i}(x) \langle u_i, x \rangle$. Therefore $\mathbb{E} \max_i \langle u_i, G \rangle \geq \sum_i \|b_i\| \geq (k/h_k) \sum_i \|b_i\|^2$, proving (10). $\square$

Lemma 4 (integrated uniform spherical-cap tail). Suppose $d \geq 3$, $s \in (0, \sqrt{d})$, and

$$B_{d,k}(s) = \frac{k}{s\sqrt{2\pi(1-1/d)}} \left(1 - \frac{s^2}{d}\right)^{(d-1)/2} \leq 1. \quad (11)$$

For all unit directions $u_1, \dots, u_k$,

$$\mathbb{E} \max_i \langle u_i, G \rangle \leq s + \frac{d}{(d-1)s}. \quad (12)$$

Proof. Write $G = RV$, with independent $R = \|G\|$ and uniform spherical direction V . By gamma log-convexity, the density of V_1 is $c_d(1-u^2)^{(d-3)/2}$ with $c_d \leq \sqrt{(d-1)/(2\pi)}$. Integrating the cap density by using $u/a \geq 1$ for $u \geq a$, gives

$$\Pr(V_1 \geq a) \leq \frac{(1-a^2)^{(d-1)/2}}{a\sqrt{2\pi(d-1)}}.$$

Union bound over the k directions and differentiate the logarithm of the right side: its negative derivative is at least $(d-1)a$ on $[a, 1)$. Hence the expected spherical maximum $T = \max_i \langle u_i, V \rangle$ satisfies $\mathbb{E}T \leq a+1/((d-1)a)$ whenever the union bound at a is at most one. Since $\mathbb{E}T \geq \mathbb{E}\langle u_1, V \rangle = 0$, the inequality $\mathbb{E}R \leq \sqrt{d}$ may safely be multiplied by $\mathbb{E}T$. Use independence of R, T and $a = s/\sqrt{d}$. $\square$

Theorem 5 (constant-sharp quadratic-dimension spherical defect). Fix $0 < c < \infty$, and let d_k be any positive integer sequence with

$$\frac{d_k}{(\log k)^2} \longrightarrow c.$$

Then

$$\liminf_{k \rightarrow \infty} k \left(U_k - F_{d_k}(k) \right) \geq \frac{1}{c}. \quad (13)$$

Proof. Write $L = \log k$, $d = d_k$. For a fixed auxiliary $\eta > 0$, choose

$$s^2 = 2L - \log L - \log(4\pi) - \frac{2L^2}{d} + \eta. \quad (14)$$

Since $d \sim cL^2$, for sufficiently large k we have $0 < s^2 < d$ and $s^2 = 2L + O(\log L)$.

Taylor's formula with explicit remainder $\log(1-z) = -z - z^2/2 + O(z^3/(1-z))$ for $0 \leq z < 1$, applied to $z = s^2/d \rightarrow 0$, gives

$$\begin{aligned} \log B_{d,k}(s) &= L - \log s - \frac{1}{2} \log(2\pi(1-1/d)) + \frac{d-1}{2} \log(1-s^2/d) \\ &= L - \frac{1}{2} \log(4\pi L) - \frac{s^2}{2} - \frac{s^4}{4d} + o(1). \end{aligned} \quad (15)$$

Indeed the discarded terms are $O(s^2/d + s^4/d^2 + s^6/d^2 + 1/d) = o(1)$ because $s^2 \asymp L$ and $d \asymp L^2$. Moreover $s^4/(4d) = L^2/d + o(1)$. Substitute (14) into (15) to obtain

$$\log B_{d,k}(s) = -\eta/2 + o(1).$$

Therefore $B_{d,k}(s) < 1$ for all large k , so Lemma 4 applies. Put $H = s + d/((d-1)s)$. As $s \rightarrow \infty$ and $d \rightarrow \infty$,

$$H^2 = s^2 + 2 + o(1) = 2L - \log L - \log(4\pi) - 2L^2/d + \eta + 2 + o(1). \quad (16)$$

Lemma 1 also gives $h_k^2 = 2L - \log L - \log(4\pi) + 2 + o(1)$. Thus

$$h_k^2 - H^2 = \frac{2L^2}{d} - \eta + o(1). \quad (17)$$

Using Lemmas 3–4 and the elementary identity

$$h_k^2 - h_k H = \frac{1}{2}(h_k^2 - H^2) + \frac{1}{2}(h_k - H)^2 \geq \frac{1}{2}(h_k^2 - H^2),$$

we obtain

$$\begin{aligned} k(U_k - F_d(k)) &= h_k^2 - kF_d(k) \\ &\geq h_k^2 - h_k H \geq L^2/d - \eta/2 + o(1). \end{aligned}$$

Take the liminf, use $L^2/d \rightarrow 1/c$, and then send $\eta \downarrow 0$. This establishes (13) with no assumed optimizer existence. $\square$

Corollary 6 (asymptotically improved dimension coefficient). Fix $C \geq 0$ and define $D_C(k)$ to be the least dimension such that $F_d(k) \geq F_\infty(k) - C/k$. Then, for all sufficiently large k , the dimension is defined (by dimension saturation), and

$$\liminf_{k \rightarrow \infty} \frac{D_C(k)}{(\log k)^2} \geq \frac{1}{C + 2(1 - \gamma)}. \quad (18)$$

Proof. The earlier finite spherical-cap converse already implies

$$D_C(k) \geq \frac{L^2}{C + 16}$$

for all sufficiently large k . Hence no subsequential limit of $D_C(k)/L^2$ can be zero. If the liminf is infinite, the claim is automatic. Otherwise take a subsequence on which $D_C(k)/L^2 \rightarrow c \in (0, \infty)$. The definition and Theorem 2 give

$$\begin{aligned} k(U_k - F_{D_C(k)}(k)) &\leq k(U_k - F_\infty(k)) + C \\ &\leq 2(1 - \gamma) + C + o(1). \end{aligned}$$

Theorem 5 gives a liminf on this subsequence of at least $1/c$. Therefore $1/c \leq C + 2(1 - \gamma)$, or $c \geq 1/(C + 2(1 - \gamma))$, as required. $\square$

The estimate is strictly stronger than the elementary coefficient $1/(C + 1)$: indeed $\gamma > 1/2$, for instance because $\gamma > H_6 - \log 7 > 1/2$ by the harmonic-integral comparison, where $H_6 = 49/20$. The logarithmic endpoint $\log 7 < 39/20$ is also certified by the companion exact checker.

Scope. This is a sharp-constant asymptotic **lower obstruction derived from spherical caps**, not a proof that the true optimal dimension coefficient equals the right side of (18). There remains a potentially substantial gap to known constructive coefficients.

3. A parameterized refinement of the binary linear-code construction

The former all-integer construction had only one preselected dimensional coefficient and an error constant 92. We retain the full dependence on dimension $m = \lceil AL^2 \rceil$, $A \geq 1$, and re-optimize the exponential-tilting cut from $B_0 = \frac{1}{2} \log L + 12$ to

$$B = \frac{1}{2} \log L + 6. \quad (19)$$

This is not a numerical heuristic: every inequality is controlled below by explicit integer or rational constants.

For the dyadic case $k = 2^r$, choose independent uniform binary columns $g_1, \dots, g_m \in \mathbb{F}_2^r$. For each message $u \in \mathbb{F}_2^r$, put

$$v_u = m^{-1/2} ((-1)^{u \cdot g_1}, \dots, (-1)^{u \cdot g_m}).$$

For a fixed generator, these are unit Gaussian score directions. A full-rank generator makes them distinct, and the diagonal sign-flip group $D_a v_u = v_{u+a}$ acts transitively on the score-maximizing Gaussian cells, so all k masses are exactly $1/k$.

For $G \sim N(0, I_m)$, let $M = \max_u \langle v_u, G \rangle$, let $a_j = G_j / \sqrt{m}$, and set

$$Q_2 = \sum_j a_j^2, \quad Q_4 = \sum_j a_j^4, \quad G = \{1/2 \leq Q_2 \leq 2, Q_4 \leq 10/m\}.$$

The Gaussian fourth and eighth moments give

$$\begin{aligned}\mathbb{E}Q_2 &= 1, & \text{Var}(Q_2) &= 2/m, \\ \mathbb{E}Q_4 &= 3/m, & \text{Var}(Q_4) &= 96/m^3,\end{aligned}$$

and Chebyshev gives $\Pr(G^c) \leq 10/m$.

Lemma 7 (uniform tilted anti-concentration, improved cut). For every $A \geq 1$, $L = \log k \geq 10^{10}$ and $m = \lceil AL^2 \rceil$, on every $G \in \mathcal{G}$ put

$$\lambda = \sqrt{2L/Q_2}, \quad \Lambda = \sum_j \log \cosh(\lambda a_j), \quad \mu = \sum_j a_j \tanh(\lambda a_j), \quad s_y = \mu - \frac{B+y}{\lambda} \quad (0 \leq y \leq \lambda/4).$$

Let $S_a = \sum_j a_j \varepsilon_j$ with independent uniform signs. Then

$$\Pr_\varepsilon(S_a \geq s_y) \geq \frac{e^{-L+B+y-1}}{20\lambda}, \quad (20)$$

and, conditional on G , the random binary-generator maximum satisfies

$$\Pr_g(M < s_y \mid G) \leq \eta e^{-y}, \quad \eta = 80e^{-5} < 3/5. \quad (21)$$

Proof. Since $Q_2 \in [1/2, 2]$, $\sqrt{L} \leq \lambda \leq 2\sqrt{L}$. Under the exponentially tilted independent-sign measure

$$\Pr_\lambda(\varepsilon_j = e) = \frac{e^{\lambda a_j e}}{2 \cosh(\lambda a_j)} \quad (e \in \{-1, 1\}),$$

the sum S_a has mean μ and variance

$$V = \sum_j a_j^2 \text{sech}^2(\lambda a_j) \in [1/4, 2],$$

because $\text{sech}^2 x \geq 1 - x^2$ and $\lambda^2 Q_4 \leq 40L/m \leq 1/4$. Every centered summand is bounded in absolute value by $2|a_j|$, so

$$\sum_j \mathbb{E}_\lambda |W_j|^3 \leq 8\sqrt{Q_2 Q_4} \leq 8\sqrt{20/m}.$$

The classical Berry–Esseen inequality for independent, not necessarily identically distributed summands with absolute constant at most 1 therefore bounds the normal CDF error by $64\sqrt{20/m} < 288/\sqrt{m}$.

Since $L \geq 10^{10}$, $B = (\log L)/2 + 6 \leq \sqrt{L}/4 \leq \lambda/4$. The first inequality follows because $\sqrt{L}/4 - (\log L)/2 - 6$ is increasing for $L \geq 16$ and is positive at 10^{10} ($\log(10^{10}) < 24$, $\sqrt{10^{10}} = 10^5$). Thus the tilted interval

$$[s_y, s_y + 1/\lambda]$$

lies, upon subtracting μ and dividing by $\sqrt{V}$, inside $[-1, 0]$, with standardized length at least $1/(2\lambda)$. Since $\varphi(1) > 1/5$, its normal probability is at least $1/(10\lambda)$. The twice-applied Berry–Esseen error is at most $576/\sqrt{m} \leq 1/(20\lambda)$ for $L \geq 10^{10}$, $m \geq L^2$. Hence its actual tilted probability is at least $1/(20\lambda)$.

The change-of-measure identity and

$$\Lambda - \lambda\mu \geq -\lambda^2 Q_2/2 = -L$$

imply (20). For the last inequality, differentiate the even function $\log \cosh x - x \tanh x + x^2/2$: its derivative for $x > 0$ is $x \tanh^2 x \geq 0$.

For each fixed G , nonzero binary messages have identically distributed Rademacher-weighted scores, and **distinct nonzero messages have pairwise independent scores over the random columns**. Indeed

two distinct nonzero vectors over $\mathbb{F}_2$ are linearly independent, so their character signs form independent pairs. Chebyshev applied to the exceedance count gives

$$\begin{aligned}\Pr_g(M < s_y \mid G) &\leq \frac{1}{(k-1) \Pr_\varepsilon(S_a \geq s_y)} \\ &\leq \frac{20\lambda e^{L-B-y+1}}{k-1} \\ &\leq 80\sqrt{L} e^{-B-y+1} = 80e^{-5-y} = \eta e^{-y}.\end{aligned}$$

Here $k-1 \geq k/2$, $\lambda \leq 2\sqrt{L}$, and $e^B = e^6\sqrt{L}$, as required. Finally $e > 27/10$ gives $e^5 > (27/10)^5 > 400/3$, so $\eta < 3/5$. $\square$

Lemma 8 (sharp dimension-dependent Gaussian maximum estimate). For every $A \geq 1$ and dyadic $k = 2^r$ with $L \geq 10^{10}$, there exists a **full-rank** binary generator of length $m = \lceil AL^2 \rceil$ such that

$$\mathbb{E}_G M \geq \sqrt{2L} - \frac{\log L}{2\sqrt{2L}} - \frac{T_A}{\sqrt{L}}, \quad T_A = 5 + \frac{10}{\sqrt{A}} + \frac{8}{A}. \quad (22)$$

The associated score-max Gaussian partition has exactly equal masses and

$$U_k - P_{\text{code}} \leq \frac{3 + 2\sqrt{2}T_A}{k}. \quad (23)$$

Proof. Conditional on $G \in G$, use the threshold failure bound from Lemma 7 and integrate downward over $s_y = s_0 - y/\lambda$. Since $M \geq X_0$, where $X_0 = m^{-1/2} \sum_j G_j \sim N(0, 1)$, the exact positive-part decomposition yields

$$\mathbb{E}_g[M \mid G] \geq \mu - \frac{B}{\lambda} - \frac{\eta}{\lambda} - \eta(4\sqrt{L} + |X_0|)e^{-\lambda/4}. \quad (24)$$

For clarity, the decomposition is

$$(s_0 - M)_+ = \frac{1}{\lambda} \int_0^{\lambda/4} \mathbf{1}_{\{M < s_y\}} dy + (s_{\lambda/4} - M)_+.$$

The remaining tail term is bounded by $(4\sqrt{L} + |X_0|)\mathbf{1}_{\{M < s_{\lambda/4}\}}$ because $\mu \leq \lambda Q_2 \leq 4\sqrt{L}$.

Let $\sigma = \sqrt{Q_2}$. Since $x \tanh(\lambda x) \geq \lambda x^2 - (\lambda^3/3)x^4$ for all real x (apply $\tanh u \geq u - u^3/3$ to $|u|$),

$$\mu \geq \sqrt{2L}\sigma - \frac{8}{3}L^{3/2}Q_4 \quad \text{on } G.$$

Using $\mathbb{E}Q_4 = 3/m$, Cauchy-Schwarz and Chebyshev give

$$\begin{aligned}\mathbb{E}[\sigma \mathbf{1}_G] &\geq 1 - \sqrt{2/m} - \sqrt{10/m}, \\ \mathbb{E}[\mu \mathbf{1}_G] &\geq \sqrt{2L} - \frac{7/\sqrt{A} + 8/A}{\sqrt{L}}.\end{aligned} \quad (25)$$

The first line follows from $|\sqrt{x} - 1| \leq |x - 1|$, and $\mathbb{E}[\sigma \mathbf{1}_{G^c}] \leq \sqrt{\mathbb{E}\sigma^2 \Pr(G^c)}$. We have

$$\mathbb{E}[(B/\lambda)\mathbf{1}_G] \leq \frac{\log L}{2\sqrt{2L}} + \frac{6}{\sqrt{2L}}, \quad \eta/\lambda < \frac{3}{5\sqrt{L}}.$$

Because $L \geq 10^{10}$, $\mathbb{E}|X_0| \leq 1$, $\eta < 3/5$, and $e^{\sqrt{L}/4} \geq (\sqrt{L}/4)^4/4!$,

$$\begin{aligned}\eta(4\sqrt{L} + \mathbb{E}|X_0|)e^{-\sqrt{L}/4} &\leq \frac{3}{5}(4\sqrt{L} + 1)\frac{6144}{L^2} \\ &\leq \frac{18432}{L^{3/2}} \leq \frac{1}{10\sqrt{L}}.\end{aligned} \quad (26)$$

Finally $6/\sqrt{2} < 17/4$, so the three dimension-independent error terms in (24) sum to

$$\frac{17/4 + 3/5 + 1/10}{\sqrt{L}} = \frac{99/20}{\sqrt{L}} < \frac{5}{\sqrt{L}}. \quad (27)$$

On the complement of G , $M \geq X_0$, and

$$\mathbb{E}[M1_{G^c}] \geq -\sqrt{10/m} \geq -\frac{1}{\sqrt{A}\sqrt{L}}$$

for $L \geq 10$. Combining (24)–(27), we get the unconditional generator-data estimate

$$\mathbb{E}_{g,G} M \geq \sqrt{2L} - \frac{\log L}{2\sqrt{2L}} - \frac{5 + 8/\sqrt{A} + 8/A}{\sqrt{L}}. \quad (28)$$

A generator has deficient rank with probability at most

$$\Pr(\text{rank} < r) \leq (k-1)2^{-m} \leq e^{L-(2/3)AL^2} \leq \frac{1}{AL^2}$$

for $L \geq 10^{10}$, $A \geq 1$. For the last estimate use $\log A \leq A-1$, $\log L \leq L$:

$$(2/3)AL^2 - L - 2\log L - \log A \geq (2/3)L^2 - 3L + [(2/3)L^2 - 1](A-1) > 0.$$

For every fixed generator,

$$\mathbb{E}_G M^2 \leq 2L + 3$$

by the Gaussian union bound on positive maxima, integration of the upper tail of M_+^2 , and $M_- \leq (X_0)_-$. Thus Cauchy–Schwarz bounds the contribution of rank-deficient generators to the average by

$$\sqrt{\frac{2L+3}{AL^2}} \leq \frac{2}{\sqrt{A}\sqrt{L}}.$$

Discard the deficient generators and normalize by the probability of the full-rank event. Since the positive lower bound for the retained average only increases upon this normalization, some full-rank generator has expected maximum satisfying (22).

Its sign-flip orbit gives exactly equal cell masses as explained above. The same Cauchy–Schwarz calculation as in the published quadratic theorem gives $kP_{\text{code}} \geq (EM)^2$. The right side of (22) is positive for our large L . Squaring gives

$$kP_{\text{code}} \geq 2L - \log L - 2\sqrt{2}T_A.$$

Mills’ normal-tail bounds imply $h_k^2 \leq 2L - \log L + 3$. Subtracting proves (23). $\square$

Technical note on constants. The use of the explicit parameter A requires *no* change to the Berry–Esseen absolute constant 1. Increasing A improves both the Gaussian weight-concentration terms $A^{-1/2}$ and the fourth-cumulant term A^{-1} . The additive floor 5 in (22) originates from the local tilted interval’s logarithmic prefactor and the integrated failure term, not from an unaccounted numerical optimizer.

4. Removing dyadic cardinality: one-coordinate exact-mass gluing

To handle every integer k , use its binary expansion, listed in strictly decreasing order:

$$k = q_1 + \dots + q_s, \quad q_j = 2^{r_j}, \quad r_1 > \dots > r_s \geq 0, \quad w_j = q_j/k.$$

The binary entropy of these selector weights satisfies the **sharp universal inequality**

$$H(w) = \sum_j w_j \log(1/w_j) < 2 \log 2. \quad (29)$$

For completeness, put $S_j = \sum_{\ell \geq j} w_\ell$. The sum of the smaller distinct powers is less than the present power, so $S_{j+1} < w_j$ and $S_j \leq 2^{1-j}$. The exact entropy chain formula

$$H(w) = \sum_j S_j h_2(w_j/S_j), \quad h_2(t) = -t \log t - (1-t) \log(1-t),$$

and $h_2 \leq \log 2$ give (29). The constant $2 \log 2$ is asymptotically attained by the binary expansion of $k = 2^s - 1$, showing that no smaller universal entropy constant is possible.

Lemma 9 (Gaussian selector and energy identity). Suppose $k = \sum_j q_j$, and for each j an equiprobable Gaussian q_j -partition in a common $\mathbb{R}^m$ has objective P_j . There is an exactly equiprobable k -partition in $\mathbb{R}^{m+1}$ with objective

$$P_{\text{glue}} \geq \frac{1}{k^2} \sum_j q_j^2 P_j. \quad (30)$$

Proof. Use one independent standard Gaussian scalar to select block j with probability q_j/k , via consecutive exact Gaussian quantiles. On the data coordinates, apply that block's q_j -cell partition. Every final cell has mass $(q_j/k)(1/q_j) = 1/k$. By independence its data-coordinate first moment is (q_j/k) times the corresponding local first moment. The sum of squared data-coordinate moments is exactly the right side of (30). The extra selector coordinate contributes nonnegative squared moments. $\square$

Theorem 10 (complete quantitative dimension-accuracy curve). Let $A \geq 1$ be any fixed real constant. For every sufficiently large integer k , there exists a measurable, exactly equiprobable k -cell Gaussian partition in dimension

$$d \leq \lceil A(\log k)^2 \rceil + 1$$

such that

$$U_k - P(\mathcal{A}) \leq \frac{\mathcal{C}(A)}{k}, \quad \mathcal{C}(A) = 4 + 4 \log 2 + 2\sqrt{2} \left(5 + \frac{10}{\sqrt{A}} + \frac{8}{A} \right). \quad (31)$$

Consequently,

$$D_{\mathcal{C}(A)}(k) \leq \lceil A(\log k)^2 \rceil + 1. \quad (32)$$

All cardinalities are included; the conclusion does not rely on a group acting transitively on *all* k cell labels, only on each dyadic subblock.

Proof. Choose the integer threshold r_0 so every dyadic $q \geq Q_0 := 2^{r_0}$ satisfies $\log q \geq 10^{10}$. Let $k \geq Q_0^2$, put $L = \log k$, and allocate a common data Gaussian block of dimension $m = \lceil AL^2 \rceil$.

For each binary block $q_j \geq Q_0$, apply Lemma 8 in its own $\lceil A(\log q_j)^2 \rceil$ -dimensional data space, then extend cylindrically to the common m dimensions. This produces an exact equal-mass q_j -partition of objective

$$P_j \geq \frac{h(1/q_j)^2 - [3 + 2\sqrt{2} T_A]}{q_j}, \quad T_A = 5 + 10/\sqrt{A} + 8/A. \quad (33)$$

For the finitely many smaller powers $q_j < Q_0$, choose any equiprobable partition, and use only $P_j \geq 0$.

Apply Lemma 9. Write $W_{\text{sm}} = \sum_{j: q_j < Q_0} w_j$. The distinct binary powers smaller than Q_0 sum to at most $Q_0 - 1$, so $W_{\text{sm}} < Q_0/k$. The squared Gaussian upper-tail hazard $h(q)^2$ is globally 2-Lipschitz in $\log(1/q)$: for every $q_j \leq k$,

$$0 \leq h(1/k)^2 - h(1/q_j)^2 \leq 2 \log(k/q_j) = 2 \log(1/w_j).$$

The identity also holds at $q_j = 1$, with $h(1) = 0$ by continuity. Therefore

$$\begin{aligned} k[U_k - P_{\text{glue}}] &\leq [3 + 2\sqrt{2} T_A] + 2 \sum_j w_j \log(1/w_j) + W_{\text{sm}} h_k^2 \\ &< 3 + 2\sqrt{2} T_A + 4 \log 2 + \frac{2Q_0 \log k}{k}. \end{aligned} \quad (34)$$

The bound $h_k^2 \leq 2 \log k$ follows from the standard Gaussian exponential-moment inequality $qe^{\lambda h(q)} \leq e^{\lambda^2/2}$, taking $\lambda = h(q)$. For $k \geq Q_0^2$,

$$\frac{2Q_0 \log k}{k} \leq \frac{4 \log Q_0}{Q_0} < 1,$$

since $Q_0 \geq 2^{10}$. Equation (34) is therefore strictly less than $\mathcal{C}(A)$, proving (31). The global optimum is no greater than the halfspace envelope U_k , so (32) follows. $\square$

Corollary 11 (seven certified explicit tradeoff points). For all sufficiently large integers k , the following bounds all hold:

Dimensional coefficient A	Guaranteed error C/k
1	$72/k$
4	$41/k$
16	$30/k$
64	$25/k$
256	$23/k$
1024	$22/k$
262144	$21/k$

In each row $d \leq \lceil A(\log k)^2 \rceil + 1$. For example,

$$\frac{1}{22 + 2(1 - \gamma)} \leq \liminf_{k \rightarrow \infty} \frac{D_{22}(k)}{(\log k)^2} \leq \limsup_{k \rightarrow \infty} \frac{D_{22}(k)}{(\log k)^2} \leq 1024. \quad (35)$$

Proof. Using the exact rational estimates

$$\log 2 < 7/10, \quad \sqrt{2} < 99/70$$

(the first follows by the degree-three Taylor lower bound for $e^{7/10} > 2$; the second by $99^2 > 2 \cdot 70^2$), we have

$$\mathcal{C}(A) < \frac{34}{5} + \frac{99}{35} \left(5 + \frac{10}{\sqrt{A}} + \frac{8}{A} \right). \quad (36)$$

For the seven square values of A in the table, the last expression is respectively below the displayed integer constants. All comparisons reduce to integer arithmetic and are independently replayed in `check_exact.py`. The first inequality in (35) is Corollary 6 with $C = 22$, and the last inequality is Theorem 10 with $A = 1024$. $\square$

The limiting guaranteed additive constant produced by this *particular* two-moment/tilting method is

$$\lim_{A \rightarrow \infty} \mathcal{C}(A) = 4 + 4 \log 2 + 10\sqrt{2}. \quad (37)$$

This number is **not** claimed to be a fundamental Gaussian quantization lower limit. The optimizing partition may have much smaller loss; our analysis is intentionally explicit and solver-free.

5. Scope, dependencies, and open quantitative questions

The new lower bound (18) and constructive upper bound (31) give a mathematically explicit Pareto-type dimension–accuracy region. They establish a stronger range of the Gaussian dimension-rate frontier than the earlier single D_{92} comparison.

The new results remain **classical mathematical proofs**, not Lean-certified theorems. The proof of the binary construction uses the standard independent-summand Berry–Esseen theorem with deliberately relaxed absolute constant 1; a published constant below 0.56 is known. The normal maximum lemma uses

elementary Mills tails and a complete uniform-integrability argument, rather than numerical extreme-value fits.

The sharp leading coefficient of $D_C(k)/(\log k)^2$ for a fixed additive C remains open. So does the existence and value of $\lim_{k \rightarrow \infty} k(U_k - F_\infty(k))$, which may be smaller than the simplex comparator $2(1-\gamma)$; no Standard Simplex global optimality is assumed.

Further directions include identifying the optimal constant in the dimension-normalized spherical defect (whether the factor $1/c$ in (13) is sharp), deriving matching upper asymptotics for fixed $d/L^2 \rightarrow c$, and producing polynomial-time certified binary generator matrices. Unequal mass vectors and the exact fixed- k Gaussian partition classification are not resolved here.

Research context and a historically cautious source screen appear in LITERATURE.md. The checker establishes exact rational parameter comparisons and a finite set of code-character, entropy, and Mills interfaces; **the universal theorems are analytic, not inferred from finite tests.**